

Teorema de Riesz-Fischer

Teorema 1 (de Riesz-Fischer). *Sea H un espacio de Hilbert y $\{u_n\}_{n \in \mathbb{N}}$ un sistema ortonormal. Entonces, la transformada de Fourier asociada a $\{u_n\}_{n \in \mathbb{N}}$ es sobreyectiva.*

Demostración: Sea $\bar{a} = (a_n)_{n \in \mathbb{N}} \in \ell^2$, definimos $\forall m \in \mathbb{N} : x_m = \sum_{n=1}^m a_n u_n$. Entonces, $(x_m)_{m \in \mathbb{N}}$ es una sucesión de Cauchy, pues para $k \in \mathbb{N}$:

$$\forall m \in \mathbb{N} : \|x_{m+k} - x_m\|^2 = \left\| \sum_{n=m+1}^{m+k} a_n u_n \right\|^2 \stackrel{\text{Pitágoras}}{=} \sum_{m+1 \leq n \leq m+k} \|a_n u_n\|^2 = \sum_{n=m+1}^{m+k} |a_n|^2 \xrightarrow{m \rightarrow \infty} 0,$$

porque $\bar{a} \in \ell^2$. Como H es completo, existe $x \in H$ tal que $\lim_m x_m = x$.

$$\forall n \in \mathbb{N} : \hat{x}_n = \langle x, u_n \rangle = \left\langle \lim_{m \rightarrow \infty} x_m, u_n \right\rangle = \lim_{m \rightarrow \infty} \langle x_m, u_n \rangle = a_n,$$

ya que $\langle x_m, u_n \rangle = a_n$ si $m \geq n$ y 0 en caso contrario. Por tanto, $\mathcal{F}(x) = \bar{a}$. ■

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.